

PCR Mid-Term Examination

Name: _____

Time: 60 Minutes

Part I: Multiple Choice Questions (10 Marks)

1. **Who is the scientist who invented the PCR technique in 1983?**

- a) James Watson
- b) Kary Mullis
- c) Francis Crick
- d) Rosalind Franklin

2. **Which enzyme is responsible for synthesizing new DNA strands in PCR?**

- a) DNA Ligase
- b) RNA Polymerase
- c) Taq Polymerase
- d) Helicase

3. **The correct sequence of steps in a single PCR cycle is:**

- a) Annealing -> Denaturation -> Extension
- b) Extension -> Annealing -> Denaturation
- c) Denaturation -> Extension -> Annealing
- d) Denaturation -> Annealing -> Extension

4. **At which temperature does the "Annealing" step typically occur?**

- a) 95°C
- b) 72°C
- c) 55°C
- d) 4°C

5. **What is the function of dNTPs in the PCR reaction?**

- a) To act as a template
- b) To serve as building blocks for the new DNA strand

- c) To provide a specific pH environment
- d) To mark the start and end of the target sequence

6. Which type of PCR is used to amplify and quantify DNA in "real-time"?

- a) Conventional PCR
- b) Nested PCR
- c) qPCR
- d) Hot Start PCR

7. In RT-PCR, what is the first step before DNA amplification can occur?

- a) DNA denaturation
- b) Conversion of RNA to cDNA
- c) Primer annealing
- d) Gel electrophoresis

8. The "Heated Lid" of the thermal cycler is designed to:

- a) Speed up the ramping rate
- b) Prevent sample evaporation
- c) Increase the cooling speed
- d) Activate the Taq polymerase

9. Which PCR variation is best suited for detecting rare mutations by dividing the sample into thousands of droplets?

- a) Multiplex PCR
- b) Nested PCR
- c) Digital PCR (dPCR)
- d) Conventional PCR

10. After 3 PCR cycles, how many copies of the target DNA are produced starting from 1 copy?

- a) 4
 - b) 6
 - c) 8
 - d) 16
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Part II: True or False (10 Marks)

- ☐ PCR stands for Polymerase Chain Reaction.
 - ☐ Magnesium ions (Mg^{2+}) are required as a cofactor for DNA polymerase activity.
 - ☐ Taq polymerase is used because it is easily destroyed by high heat.
 - ☐ In qPCR, a lower Ct value indicates a higher starting amount of target DNA.
 - ☐ Primers are short synthetic DNA sequences, typically 18-22 bases long.
 - ☐ Conventional PCR allows for the quantification of DNA while the reaction is running.
 - ☐ Multiplex PCR uses only one pair of primers to amplify multiple targets.
 - ☐ The "Hold" step at 4°C is used to safely store samples after the run is finished.
 - ☐ Denaturation involves breaking hydrogen bonds between DNA strands at roughly 95°C.
 - ☐ Nested PCR is used to decrease the specificity of the reaction.
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Part III: Fill in the Blanks (10 Marks)

1. The instrument used to perform PCR by rapidly changing temperatures is called a _____.
 2. _____ is a technique used to study gene expression by analyzing mRNA levels.
 3. The _____ rate refers to how fast the machine can heat or cool between steps.
 4. To prevent non-specific amplification at room temperature, scientists use _____ PCR.
 5. In the extension step, the polymerase adds nucleotides in the _____ direction.
 6. The _____ value is the cycle number where fluorescence crosses a specific threshold.
 7. RT-PCR requires an enzyme called _____ to convert RNA into cDNA.
 8. _____ PCR improves specificity by using two rounds of PCR with two sets of primers.
 9. The _____ solution maintains the optimal pH and salt conditions for the reaction.
 10. If you start with 1 copy of DNA, after 20 cycles you will have approximately _____ copies.
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Part IV: Applied Analysis Question

Refer to the qPCR Amplification Plot (Fluorescence vs. Cycle Number) and answer:

1. **Compare Sample A (Ct = 18) and Sample B (Ct = 28):**
 - Which sample had a higher initial concentration of DNA? Why?

2. What does the "Threshold" line represent in this graph?

Answer Key

- **Part I (MCQ):** 1-b, 2-c, 3-d, 4-c, 5-b, 6-c, 7-b, 8-b, 9-c, 10-c
- **Part II (T/F):** 1-T, 2-T, 3-F, 4-T, 5-T, 6-F, 7-F, 8-T, 9-T, 10-F
- **Part III (Fill):** 1-Thermal Cycler, 2-RT-PCR, 3-Ramping, 4-Hot Start, 5-5' to 3', 6-Ct, 7-Reverse Transcriptase, 8-Nested, 9-Buffer, 10-1 Million
- **Part IV:** 1-Sample A (Earlier detection/Lower Ct). 2-The point where signal exceeds background noise.